E59: Idempotence and Orthogonality of the Dominant-Annulus Fourier Projection

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Abstract

We show that restriction to a fixed dominant Fourier annulus on the periodic domain is an idempotent, self-adjoint contraction on L^2, hence an orthogonal projection. We also establish its exact energy decomposition, its best-approximation property, and the corresponding discrete Fourier transform identity. The result concerns a fixed sharp spectral mask; it does not assert invariance of the selected subspace under nonlinear evolution.

1 Definitions and theorem

Let $d \geq 1$, let $\mathbb{T}^d = (\mathbb{R}/2\pi\mathbb{Z})^d$, and equip $L^2(\mathbb{T}^d;\mathbb{C})$ with the normalized inner product

$$\langle A, B\rangle := \frac{1}{(2\pi)^d}\int_{\mathbb{T}^d} A(x)\overline{B(x)}\,dx, \qquad \|A\|_2^2 = \langle A, A\rangle. \tag{1}$$

Its Fourier coefficients are

$$\widehat{A}_k = \frac{1}{(2\pi)^d}\int_{\mathbb{T}^d} A(x)e^{-\mathrm{i}k\cdot x}\,dx, \qquad k \in \mathbb{Z}^d, \tag{2}$$

and $A = \sum_{k\in\mathbb{Z}^d} \widehat{A}_k e^{\mathrm{i}k\cdot x}$ with convergence in L^2. Parseval's identity states

$$\langle A, B\rangle = \sum_{k\in\mathbb{Z}^d} \widehat{A}_k\overline{\widehat{B}_k}, \qquad \|A\|_2^2 = \sum_{k\in\mathbb{Z}^d} |\widehat{A}_k|^2. \tag{3}$$

Fix a preferred wavenumber $k_\star \geq 0$ and bandwidth $\Delta k \geq 0$. Define the *fixed* dominant annulus

$$\mathcal{A}^\star := \left\{k \in \mathbb{Z}^d : ||k| - k_\star| \leq \Delta k\right\}. \tag{4}$$

Let $\chi_{\mathcal{A}^\star}$ be its indicator, and set

$$(P_{k_\star}A)(x) := \sum_{k\in\mathcal{A}^\star} \widehat{A}_k e^{\mathrm{i}k\cdot x} = \sum_{k\in\mathbb{Z}^d} \chi_{\mathcal{A}^\star}(k)\widehat{A}_k e^{\mathrm{i}k\cdot x}. \tag{5}$$

The corresponding band-limited subspace is

$$\mathcal{H}_{\mathcal{A}^\star} := \left\{B \in L^2(\mathbb{T}^d) : \widehat{B}_k = 0 \text{ for all } k \notin \mathcal{A}^\star\right\}. \tag{6}$$

Theorem 1 (E59: dominant-annulus projection)**.** *The operator $P_{k_\star} : L^2(\mathbb{T}^d) \to L^2(\mathbb{T}^d)$ defined by* (5) *is bounded and satisfies*

$$\boxed{P_{k_\star}^2 = P_{k_\star}, \qquad P_{k_\star}^* = P_{k_\star}, \qquad \|P_{k_\star} A\|_2 \le \|A\|_2 \quad (A \in L^2).} \tag{7}$$

It is the orthogonal projection onto $\mathcal{H}_{\mathcal{A}^\star}$, with

$$\boxed{\|A\|_2^2 = \|P_{k_\star} A\|_2^2 + \|(I - P_{k_\star})A\|_2^2.} \tag{8}$$

Furthermore, $P_{k_\star} A$ is the unique best L^2 approximation to A from $\mathcal{H}_{\mathcal{A}^\star}$:

$$\boxed{P_{k_\star} A = \operatorname*{arg\,min}_{B \in \mathcal{H}_{\mathcal{A}^\star}} \|A - B\|_2.} \tag{9}$$

If $\mathcal{A}^\star \neq \varnothing$, then $\|P_{k_\star}\|_{\mathrm{op}} = 1$; if $\mathcal{A}^\star = \varnothing$, then $P_{k_\star} = 0$.

Proof. **Well-definedness and boundedness.** Because $\chi_{\mathcal{A}^\star}(k) \in \{0, 1\}$, the coefficient sequence $(\chi_{\mathcal{A}^\star}(k)\widehat{A}_k)_{k \in \mathbb{Z}^d}$ belongs to ℓ^2. Thus (5) defines an L^2 function. By Parseval,

$$\|P_{k_\star} A\|_2^2 = \sum_{k \in \mathbb{Z}^d} \chi_{\mathcal{A}^\star}(k)^2 |\widehat{A}_k|^2 = \sum_{k \in \mathcal{A}^\star} |\widehat{A}_k|^2 \le \sum_{k \in \mathbb{Z}^d} |\widehat{A}_k|^2 = \|A\|_2^2. \tag{10}$$

In particular $P_{k_\star}$ is a bounded linear map.

Idempotence. Uniqueness of Fourier coefficients gives

$$\widehat{(P_{k_\star} A)}_k = \chi_{\mathcal{A}^\star}(k)\widehat{A}_k. \tag{11}$$

Applying $P_{k_\star}$ once more,

$$P_{k_\star}^2 A = \sum_{k \in \mathbb{Z}^d} \chi_{\mathcal{A}^\star}(k) \widehat{(P_{k_\star} A)}_k e^{\mathrm{i}k\cdot x} \tag{12}$$

$$= \sum_{k \in \mathbb{Z}^d} \chi_{\mathcal{A}^\star}(k)^2 \widehat{A}_k e^{\mathrm{i}k\cdot x} \tag{13}$$

$$= \sum_{k \in \mathbb{Z}^d} \chi_{\mathcal{A}^\star}(k) \widehat{A}_k e^{\mathrm{i}k\cdot x} = P_{k_\star} A. \tag{14}$$

The penultimate equality uses precisely $\chi_{\mathcal{A}^\star}(k)^2 = \chi_{\mathcal{A}^\star}(k)$ for every k. Since A is arbitrary, $\boxed{P_{k_\star}^2 = P_{k_\star}}$.

Self-adjointness and range. For $A, B \in L^2$, Parseval yields

$$\langle P_{k_\star} A, B \rangle = \sum_k \chi_{\mathcal{A}^\star}(k) \widehat{A}_k \overline{\widehat{B}_k} \tag{15}$$

$$= \sum_k \widehat{A}_k \overline{\chi_{\mathcal{A}^\star}(k) \widehat{B}_k} = \langle A, P_{k_\star} B \rangle, \tag{16}$$

because $\chi_{\mathcal{A}^\star}$ is real. Hence $P_{k_\star}^* = P_{k_\star}$. Every $P_{k_\star} A$ has Fourier support in $\mathcal{A}^\star$, so $\mathrm{Ran}(P_{k_\star}) \subseteq \mathcal{H}_{\mathcal{A}^\star}$. Conversely, if $B \in \mathcal{H}_{\mathcal{A}^\star}$, then $P_{k_\star} B = B$ by the multiplier formula, so $\mathcal{H}_{\mathcal{A}^\star} \subseteq \mathrm{Ran}(P_{k_\star})$. Thus $\mathrm{Ran}(P_{k_\star}) = \mathcal{H}_{\mathcal{A}^\star}$. Self-adjointness and idempotence show that $P_{k_\star}$ is its orthogonal projection.

Energy decomposition. The Fourier supports of $P_{k_\star}A$ and $(I-P_{k_\star})A$ are disjoint, so their inner product vanishes by Parseval: $\langle P_{k_\star}A, (I-P_{k_\star})A\rangle = 0$. Expanding $A = P_{k_\star}A + (I-P_{k_\star})A$ gives (8).

Best approximation. For any $B \in \mathcal{H}_{\mathcal{A}^\star}$, the vectors $(I-P_{k_\star})A$ and $P_{k_\star}A - B \in \mathcal{H}_{\mathcal{A}^\star}$ are orthogonal. Therefore

$$\|A-B\|_2^2 = \|(I-P_{k_\star})A\|_2^2 + \|P_{k_\star}A - B\|_2^2 \geq \|(I-P_{k_\star})A\|_2^2, \tag{17}$$

with equality exactly when $B = P_{k_\star}A$. This proves (9). Finally, (10) implies $\|P_{k_\star}\|_{\mathrm{op}} \leq 1$. If $k \in \mathcal{A}^\star$, the nonzero function $e^{\mathrm{i}k\cdot x}$ is fixed by $P_{k_\star}$, proving $\|P_{k_\star}\|_{\mathrm{op}} = 1$; when $\mathcal{A}^\star$ is empty, (5) gives the zero operator. □

2 Spectral concentration (E62)

For $A \neq 0$, define the retained spectral-energy fraction

$$\eta(A) := \frac{\sum_{k\in\mathcal{A}^\star} |\widehat{A}_k|^2}{\sum_{k\in\mathbb{Z}^d} |\widehat{A}_k|^2} = \frac{\|P_{k_\star}A\|_2^2}{\|A\|_2^2}. \tag{18}$$

Equation (10) immediately implies

$$0 \leq \eta(A) \leq 1, \qquad \eta(A) = 1 \quad \Longleftrightarrow \quad P_{k_\star}A = A. \tag{19}$$

3 Discrete FFT implementation

Let $F \in \mathbb{C}^{N\times N}$ be a unitary discrete Fourier transform matrix ($F^*F = FF^* = I$), and let $D = \mathrm{diag}(m_1, \ldots, m_N)$ with $m_j \in \{0,1\}$ represent a *fixed* discrete annulus mask. The physical-space operator $P = F^*DF$ satisfies

$$P^2 = F^*DFF^*DF = F^*D^2F = F^*DF = P, \tag{20}$$

since $D^2 = D$. Likewise $P^* = P$. Floating-point FFT implementations satisfy these identities up to rounding error when the same mask and transform conventions are used twice.

4 Scope and qualifications

Remark 1 (Correct annulus convention)**.** *The symmetric condition $\big||k| - k_\star\big| \leq \Delta k$ defines a shell about the preferred wavenumber. The one-sided condition $|k| - k_\star \leq \Delta k$ defines a different set containing low wavenumbers. Idempotence holds for a fixed mask of either kind, but only the symmetric condition encodes the intended annulus.*

Remark 2 (Smooth multipliers)**.** *For any bounded Fourier multiplier Q defined by $\widehat{QA}_k = w(k)\widehat{A}_k$, one has $Q^2 = Q$ if and only if $w(k)^2 = w(k)$ for every admissible Fourier mode k, i.e., $w(k) \in \{0,1\}$. A genuinely tapered band-pass filter is therefore not generally idempotent.*

Remark 3 (Adaptive selections and nonlinear evolution)**.** *The proof assumes that $k_\star$ and Δk are held fixed under successive applications. If a "dominant annulus" is selected afresh from each input, the resulting map need not be idempotent. Moreover, the fact that $P_{k_\star}$ is a projection does not imply that $\mathcal{H}_{\mathcal{A}^\star}$ is invariant under an unprojected nonlinear PDE, nor does it by itself demonstrate localization, confinement, or stability.*

Conclusion

For the fixed, sharp dominant-annulus Fourier mask used in E59,

$$\boxed{P_{k_\star}^2 = P_{k_\star}, \qquad P_{k_\star}^* = P_{k_\star}, \qquad \|P_{k_\star} A\|_2 \le \|A\|_2.}$$

Thus E59 is proved as an exact Hilbert-space operator statement.